Chance-constrained anytime replanning for multi-UAV missions under uncertain execution, with pre-registered validation on PX4 and Gazebo

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Abstract

Multi-UAV replanners usually plan for nominal execution and stop their search after a fixed time budget. Under wind, speed and service-time errors, such plans promise more than the vehicles deliver. This paper proposes AnytimeSwarm-CC, a replanner that takes three decisions in every replanning cycle. It replans on events and periodically, plans against a chance constraint on the finish time of every task, and stops an anytime adaptive large neighbourhood search when a learned prediction of the next improvement no longer exceeds a delay cost. Because the mean and the spread of a calibrated finish-time error grow linearly with the lead time, the chance constraint reduces exactly to an effective deadline per task, and the fast repair and search heuristics are reused unchanged. The evaluation uses a kinematic simulator with disturbances drawn with common random numbers and PX4 software-in-the-loop with Gazebo and ROS 2, and two pre-registered studies with frozen code and held-out seeds. In 48 600 kinematic runs, the proposed replanner reached a utility of 52.15%. It served 1.35 to 2.87 percentage points more reward than six baselines, among them stopping rules from the metareasoning literature and a robust time buffer (all Holm-adjusted $p < 0.001$), 9.3% more than a conventional planner, and 49% fewer tasks late than a fixed budget of 40 ms. Ablations attribute the gain to the chance-constrained model. On this model, the learned stopping rule was equivalent to a fixed budget of 40 ms, and under high disturbances with halved time windows the model was too cautious and a constant buffer served more reward. In software-in-the-loop flights of 30 missions, the proposed replanner served 25% fewer tasks late than the fixed budget, but the utility differences stayed within the noise ($p_{\text{Holm}} \geq 0.942$), so the gain was not confirmed at this sample size.

Keywords: multi-UAV task allocation, dynamic vehicle routing, chance-constrained planning, anytime planning, metareasoning, execution uncertainty, PX4, Gazebo, ROS 2

1. Introduction

Teams of unmanned aerial vehicles (UAVs) are used for inspection, delivery and monitoring. During a mission, new tasks may appear, a vehicle may fail, or a deadline may be moved forward. A central planner then repairs the routes of the whole fleet, while some UAVs may already wait for new instructions. Dynamic vehicle routing and multi-robot task allocation offer many methods for this repair [1–3], and metaheuristics such as adaptive large neighbourhood search (ALNS) [4] usually find good plans when enough time is available.

The time given to such a planner is usually fixed in advance, and the same budget is spent on easy and on hard events. This choice may be inefficient for two reasons. Many events leave little room for improvement,

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and on the others most of the improvement of an anytime search tends to arrive early. In addition, idle UAVs wait while the planner computes, so a later plan is not always a better plan. Anytime algorithms and metareasoning study this question in general terms [5–10]. In replanning, the cost of time is specific to the current plan. It depends on which UAVs are idle, on the slack in their routes and, in practice, on how reliably the plan will be executed.

Execution uncertainty changes the problem in two ways. First, a planner that assumes exact execution accepts plans whose finish times leave no margin. When UAVs meet wind, fly slower than commanded or take longer at a task, part of the promised slack is consumed by errors that the planner does not see, and tasks near the end of a route are served late. The error also grows with the time until a task is reached, so a fixed margin is too large for early tasks and too small for late ones. Second, the cost of a later plan depends on how reliably the plan will be executed, and errors accumulate between two replanning calls, which raises the question of when to replan at all. Stochastic and robust routing protect schedules against such errors [11, 12], but usually with models that are too slow to be solved within the fraction of a second available between events. Validation is a further difficulty. Results obtained with a kinematic simulator may not hold for vehicles flown by an autopilot, while a physics-based simulation alone cannot provide the number of missions that a statistical comparison of planners requires.

This paper proposes AnytimeSwarm-CC, a replanner that addresses these points together, and evaluates it against baselines from the literature. Its contributions are fourfold. The first is a chance-constrained planning model for fast replanning. A calibrated model of the finish-time error, whose mean and spread grow linearly with the lead time, turns a chance constraint on the finish time of every task into an effective deadline in closed form. The proposed model is therefore solved by the same feasibility-first repair and anytime adaptive large neighbourhood search (ALNS) as the nominal model, with no extra cost per call. The second is a learned stopping rule. The search stops when a prediction of the next improvement no longer exceeds a delay cost derived from the slack of the plan, and an execution-aware variant prices the delay with the calibrated error model. Two propositions bound the reward that a later publication loses and show that the execution-aware cost reduces to the deterministic one when the error vanishes. The third is a two-tier evaluation framework. Tier B is a continuous-time kinematic simulator with acceleration limits, wind, speed and service-time errors and stochastic communication latency, drawn with common random numbers [13] so that every planner faces the same disturbances. Tier C flies the same missions with three PX4 autopilots [14] in Gazebo [15], connected to the planner through ROS 2 [16], after a rescaling that keeps every dimensionless ratio of the decision problem. The fourth is a pre-registered evaluation in two studies. Study 1 tests whether an execution-aware stopping rule alone improves the nominal planner. Study 2 compares the proposed replanner with re-implemented stopping rules from the metareasoning literature [8, 10, 17], size-scaled and fixed budgets, a robust time buffer [12] and a conventional event-triggered planner, over latency factors and time-window widths. All design choices were fixed on development seeds, the code was frozen with cryptographic hashes, and each study was run once on its own held-out seeds with mission-level paired bootstrap, Holm correction and equivalence tests.

The results are reported as they came out. In study 1, the execution-aware stopping rule needed 15% less computation than its deterministic counterpart, but it did not serve significantly more reward than the strongest baselines. In study 2, the proposed replanner served 1.35 to 2.87 percentage points more reward than each of six baselines and 9.3% more than the conventional planner. Ablations show that this gain comes from the chance-constrained model, not from the stopping rule, which was equivalent to a fixed budget on the same model. The proposed model was too cautious under high disturbances with halved time windows, where a constant buffer served more reward.

The remainder of this paper is organised as follows. Section 2 reviews related work. Section 3 formulates replanning with publication delay and uncertain execution. Section 4 describes the proposed replanner. Section 5 presents the evaluation framework, the compared methods and the protocol. Section 6 reports the results, Section 7 discusses them together with the limitations of the study, and Section 8 concludes the

Table 1: Features of recent approaches to dynamic task allocation and replanning compared with the proposed method. TW: time windows or deadlines; Exec. unc.: execution or duration uncertainty in the planning model; Trigger: replanning trigger policies compared; Stop: when to stop the replanner (computation as a decision); SITL: validation with an autopilot in software-in-the-loop or physics simulation. ✓: yes; ∘: partly; –: no or not reported.

Work	Problem	TW	Exec. unc.	Trigger	Stop	SITL	Architecture
Choudhury et al. [27]	MRTA, time windows	✓	✓	–	–	–	hierarchical
Huang et al. [28]	multi-UAV allocation	–	✓	∘	–	–	–
Calvo and Capitan [32]	long-endurance UAV MRTA	∘	–	–	–	✓	centralised
Hwang et al. [30]	multi-UAV replanning	✓	–	–	–	–	–
Shen et al. [31]	UAV swarm allocation	∘	–	–	–	–	edge
Qian and Fu [29]	multi-UAV allocation	✓	✓	–	–	–	distributed
Cong et al. [34]	multi-UAV allocation	✓	∘	✓	–	∘	–
Ali et al. [33]	multi-UAV allocation	–	–	–	–	✓	decentralised
Svegliato et al. [10]	anytime metareasoning	–	–	–	✓	–	–
Proposed	multi-UAV replanning	✓	✓	✓	✓	✓	centralised

paper.

2. Related work

Dynamic task allocation and routing. Multi-robot task allocation [1] and dynamic vehicle routing [2, 3, 18] study how tasks can be reassigned when information changes. The planning problem considered here is a prize-collecting routing problem with time windows [19, 20]. Real-time routing has long been limited by the computation available between events. Gendreau et al. parallelised tabu search to fit more optimisation between request arrivals [21], and large neighbourhood search has been adapted to dynamic routing with time windows [22, 23] and to routing with drones [24]. For multi-UAV systems, auction methods such as CBBA [25, 26] reach conflict-free assignments quickly. Recent work handles stochastic task completion with hierarchical policies [27], uncertain execution times with dynamic time slices [28], intermittent communication with uncertainty-gated distributed allocation [29], and events with assignment-preserving or local repair of the previous plan [30, 31]. Calvo and Capitan validate a repair-based architecture for long-endurance missions with PX4 and Gazebo [32], and Ali et al. use ArduPilot software-in-the-loop simulation [33]. Table 1 compares these approaches. They make the search faster or the plan better within a given budget, but none of them treats the computation time of the replanner as a decision, calibrates the finish-time error of the executed routes, or compares replanning triggers together with the stopping rule.

Replanning policies. Rescheduling research distinguishes event-driven, periodic and hybrid policies and studies how this choice affects the performance and the stability of the schedule [35, 36]. In dynamic and stochastic routing, rolling-horizon re-optimisation is the standard way to absorb new information [18]. For multi-UAV allocation, Cong et al. compare event-triggered with periodic allocation [34]. The present work evaluates the trigger jointly with the stopping rule and the planning model, because more frequent calls change how much computation each call should use.

Uncertainty in routing and allocation. Stochastic vehicle routing models travel and service times or demands as random variables [37]. With hard time windows and stochastic service times, a priori routes with recourse [11, 38] and robust formulations over budgeted uncertainty sets [12] protect the schedule against delays, at the price of solving a harder model. Scenario-based approaches plan against sampled futures in dynamic settings [39], and chance-constrained task allocation bounds the probability that a team misses its objectives [40]. The proposed planning model follows this line but exploits the structure of a calibrated error model: because the mean and the spread of the finish-time error grow linearly with the lead time, a chance

constraint on every task reduces exactly to an effective deadline, and the fast repair and search heuristics are reused without change.

Metareasoning and anytime algorithms. Anytime algorithms [5, 8] and the rational allocation of computation [6, 7, 17] form the conceptual basis of the stopping rule. Hansen and Zilberstein derive a monitoring policy by dynamic programming over a performance profile and a time-dependent utility [9]. Svegliato et al. predict the performance of an anytime algorithm online [10], learn meta-level control with model-free reinforcement learning [41] and tune the hyperparameters of anytime planners with deep reinforcement learning [42], and Sung et al. learn when an anytime motion planner should stop [43]. Situated temporal planning accounts for the passage of time during planning [44] and allocates search effort to actions that expire [45, 46], and deadline-aware search adapts the search to a fixed deadline from on-line measures of its progress [47]. In these studies, the utility of time is usually a given function or a hard deadline. In multi-UAV replanning, it follows from the plan at hand, from the vehicles that wait for it and from how reliably the plan will be executed. Two of these approaches, static performance profiles and online performance prediction, are re-implemented as baselines in this paper.

Simulation-based validation of multi-UAV planning. Software-in-the-loop simulation with an autopilot and a physics simulator is a common step between algorithm design and flight tests. PX4 [14] runs the same flight stack in simulation and on hardware, Gazebo [15] provides vehicle dynamics and sensors, and ROS 2 [16] connects planners and vehicles. Such setups are rarely used for statistical comparisons of planners, because each mission runs in real time or close to it. The two-tier design of this paper uses the physical tier to check a fast simulator that carries the statistical burden.

3. Problem formulation

3.1. Replanning snapshot

A central planner is called whenever a replanning trigger fires (Section 3.4). Each UAV keeps the task that it is currently flying to or serving. This committed task determines when and where the UAV becomes available again and how much energy it has left. Each open task has a location, a release time, a deadline by which its service must be completed, a service duration and a reward. A plan assigns an ordered sequence of open tasks to each UAV, and tasks may remain unassigned. The planning problem is therefore a prize-collecting routing problem with time windows [19, 20]. Arrival, start and finish times along a route are computed in the usual way for time windows [19, 48]. A route is feasible if every service ends before its deadline and the UAV returns to the depot with an energy reserve of 10% of its capacity. Flight consumes energy in proportion to the distance, hovering during service consumes energy over time, and waiting consumes no energy. A plan π is evaluated by

$$J(\pi) = \alpha R(\pi) - \gamma E(\pi) - \delta C(\pi), \tag{1}$$

where R is the reward of the tasks served on time, E is the flown distance in kilometres, and C is the number of tasks that move to a different UAV than in the previous plan. The weights are $\alpha = 1$, $\gamma = 0.05$ per kilometre and $\delta = 0.3$. Plan quality is reported in percent of W, the total reward of the open tasks.

3.2. Publication delay

Computation is not free. A plan that is computed for τ milliseconds is released $\kappa\tau$ milliseconds later in mission time. The latency factor κ converts planning time into mission time. The value $\kappa = 1$ corresponds to a planner that runs in real time on the evaluation computer, whereas larger values represent slower on-board processors. A UAV that is already idle must wait for the new plan, so its whole route starts later, and tasks

whose deadline can no longer be met are lost. With $J_{\text{eff}}(\pi, \tau)$ as the objective of π after this shift, the net utility of publishing π after τ milliseconds is

$$U(\pi, \tau) = J_{\text{eff}}(\pi, \tau) - \mu W \tau, \tag{2}$$

where μ is the price of one millisecond of computation as a fraction of W. The planner thus chooses both a plan and the moment at which it stops computing.

3.3. Uncertain execution and dissemination

The planner uses a nominal model: straight legs at the cruise speed of each UAV and nominal service durations. The executed mission differs from this model in two ways. First, the finish time of task j is $F_j = f_j + e_j$, where f_j is the nominal finish time computed at publication and e_j is an execution error caused by acceleration, wind, speed tracking and service-time variability. Second, the plan reaches the vehicles after an additional dissemination latency L that is not under the control of the planner. A task yields its reward only if its actual finish time meets its deadline, which may have been tightened in the meantime. Following common practice in dispatching, a UAV skips a task at departure if the nominal estimate from its actual state already misses the deadline, and the task returns to the pool of open tasks. The performance measure of a mission is its utility, the reward of all tasks served on time divided by the total reward of all tasks of the mission.

3.4. Replanning triggers

Three trigger policies are considered. The event trigger replans on task arrivals, UAV failures and deadline changes only. The periodic trigger additionally replans every T seconds of mission time. The deviation trigger additionally replans when the actual finish of a planned task deviates from its planned finish by more than θ seconds. During planning, every UAV finishes its committed task and then waits for the new plan, and events that arrive during planning are queued. These semantics are identical in both simulation tiers.

4. Proposed method

The proposed replanner, AnytimeSwarm-CC, decides at three points of the replanning cycle (Fig. 1). The trigger decides when to replan. The chance-constrained planning model decides which plans are acceptable given the calibrated execution error. The stopping rule decides when to stop computing and publish. Between these decisions, a feasibility-first repair and an anytime adaptive large neighbourhood search (ALNS) produce the plan. Algorithm 1 summarises one replanning call. Earlier variants of the replanner differ only in parts of this cycle: AnytimeSwarm uses the deterministic delay cost and the nominal planning model, and AnytimeSwarm-R adds the execution-aware delay cost but keeps the nominal planning model.

4.1. Feasibility-first repair

The previous plan is first restricted to the open tasks. Each route is then walked once, and a task is kept only if the route up to this task is still feasible, so the result is feasible by construction. Tasks of a failed UAV, removed tasks and new tasks are then inserted with parallel regret-2 insertion [4, 49], and only insertions that increase the objective are made. This feasibility-first repair (FFR) yields a publishable plan within milliseconds and provides the starting point of the search.

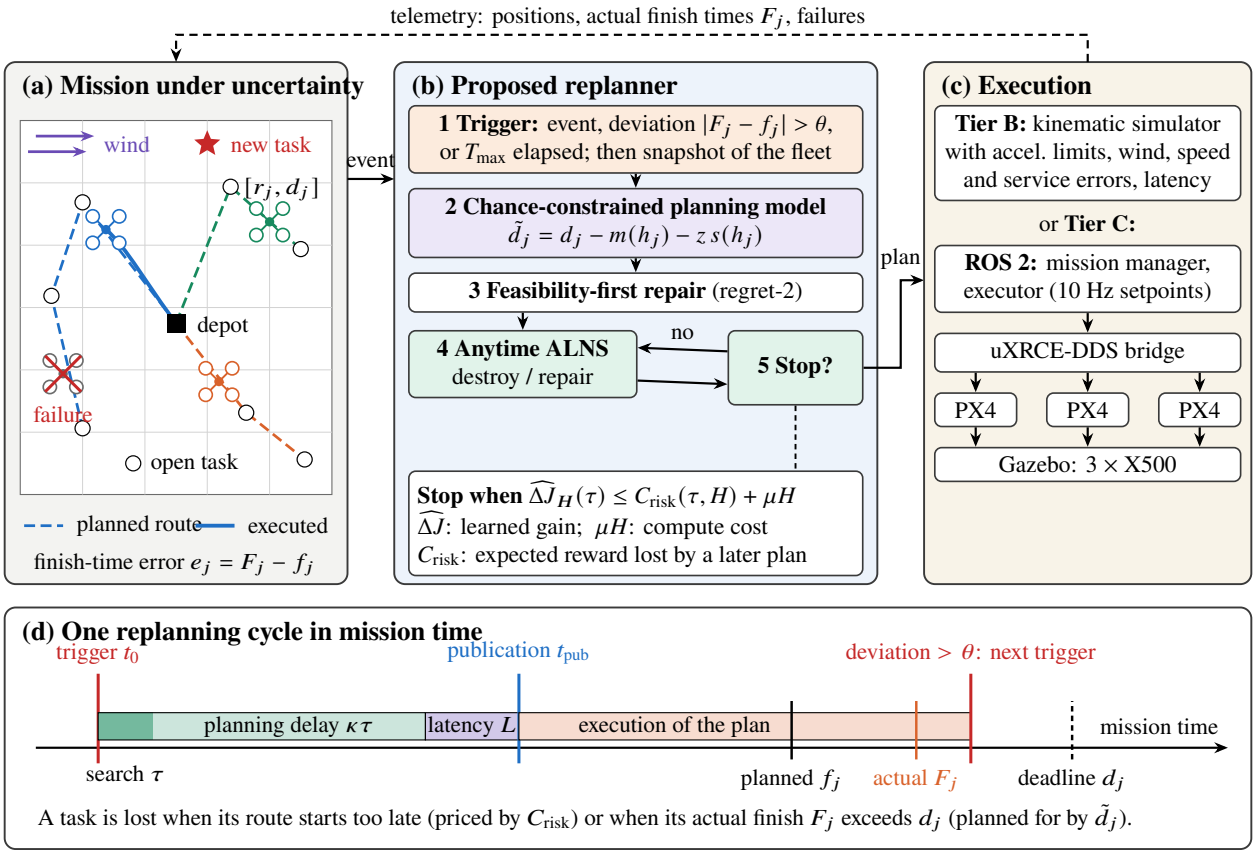


Figure 1: Overview of the proposed replanner. (a) The mission: UAVs serve tasks with time windows, while wind and speed and service errors make the executed routes deviate from the plan, and events change the task set. (b) One replanning call: trigger, chance-constrained planning model, feasibility-first repair, anytime ALNS and the stopping rule. (c) Execution in the kinematic tier or in PX4 software-in-the-loop with Gazebo and ROS 2. (d) The mission-time view of one call: search, emulated planning delay, dissemination latency and execution, whose deviations trigger the next call.

4.2. Anytime adaptive large neighbourhood search

The plan is then improved by ALNS [4]. Four destroy operators remove between 10% and 35% of the assigned tasks, and at least two. They select tasks at random, by their contribution to the objective, by relatedness [50] or as a segment of a route. Four repair operators apply greedy or regret-2 insertion, each with and without noise. Operators are chosen by adaptive roulette-wheel selection, and new solutions are accepted by simulated annealing [51]. All intermediate solutions are feasible, so the best plan found so far can be published at any time. The temperature decreases per iteration rather than per unit of time, and all variants use the same random seed. Every stopping rule therefore follows the same search trajectory and differs only in the moment at which it stops.

4.3. Learned stopping rule

Every 5 ms, the stopping rule compares the predicted gain of searching for another $H = 10$ ms with the cost of doing so. With $\widehat{\Delta J}$ as the predicted gain, the search stops at time τ if

$$\widehat{\Delta J} \;\le\; C_{\text{delay}}(\tau) + \mu W H. \tag{3}$$

The gain is predicted from the search history by a Poisson generalised linear model with a log link [52] and a small L_2 penalty. Its eleven features describe the elapsed time, the number of iterations, the recent

Algorithm 1 One replanning call of the proposed replanner

Require: snapshot at time t_0: committed tasks, open tasks with (r_j, d_j, w_j), UAV states; previous plan π^-; error model (b_0, b_1, s_0, s_1), safety factor z, expected latency $\bar{L}$, latency factor κ

$\tilde{d}_j \leftarrow \min\{d_j,\ t_0 + (d_j - t_0 - b_0 - z s_0)/(1 + b_1 + z s_1)\}$ for every open task ▷ Eq. (7)

$\pi \leftarrow \textsc{FeasibilityFirstRepair}(\pi^-, \tilde{d})$ ▷ feasible by construction

$\pi^* \leftarrow \pi;\ \tau \leftarrow 0$

repeat

  destroy and repair π with adaptively chosen operators; accept by simulated annealing

  if $J(\pi) > J(\pi^*)$ **then** $\pi^* \leftarrow \pi$

  every 5 ms: $\widehat{\Delta J} \leftarrow$ learned prediction of the gain in the next H ms

  $C \leftarrow \Lambda_R(\pi^*, \tau + H) - \Lambda_R(\pi^*, \tau)$ ▷ Eq. (5)

until $\widehat{\Delta J} \le C + \mu W H$ **or** $\tau \ge B$

publish π^*; it reaches the UAVs at $t_0 + \kappa\tau + L$

next call at the next event or $T_{\max}$ after t_0, whichever comes first

improvements over 5 ms, over 20 ms and over the second half of the elapsed time, the time since the last improvement, the total improvement over the repaired plan, the unassigned share of the reward, the numbers of tasks and UAVs, and the iteration rate. The target is the improvement in the next 10 ms in percent of W. The model is fitted on 500 ms search trajectories of development scenarios only, and a prediction costs one dot product. The search loop enforces the hard time budget B independently of the stopping rule. The price of computation is $\mu = 10^{-5}$ per millisecond.

4.4. Deterministic delay cost

The delay cost is computed from the best plan found so far. A late start of a route is partly absorbed by the waiting times along it, in the same way as the forward time slack of routing with time windows [48]. The tolerance tol_j of a task is the margin between its finish time and its deadline, plus all waiting times on the route up to this task. A task is lost when the start shift $\sigma_j(\tau)$ of its route exceeds its tolerance. The start shift is zero for a UAV that is still busy, and it equals the waiting time until the publication for an idle UAV. With $v_j = \alpha w_j - \delta\,\mathbb{1}[j \text{ changes UAV}]$ as the value of task j in the objective and $\Lambda(\tau) = \sum_j v_j\,\mathbb{1}[\sigma_j(\tau) > \mathrm{tol}_j]$, the delay cost of searching H milliseconds longer is $C_{\mathrm{delay}}(\tau) = \Lambda(\tau + H) - \Lambda(\tau)$.

Proposition 1. *For a feasible plan executed exactly, (i) $\Lambda(\tau)$ is the value lost through missed deadlines if every UAV follows its route, (ii) the loss of the objective is at most $\Lambda(\tau)$ if late tasks are skipped, and (iii) $\Lambda(\tau)$ does not decrease with τ and is zero while all UAVs are busy with their committed tasks. Hence C_{delay} is non-negative and vanishes while no UAV is waiting.*

Proof. A route that starts later by some shift reaches its first task later by the same amount. Each waiting time absorbs part of the delay, so the delay at the end of a task equals the start shift minus all earlier waiting times, if this difference is positive. The task therefore misses its deadline exactly when the start shift exceeds its tolerance. Waiting consumes no energy, which proves (i). If late tasks are skipped, the triangle inequality ensures that every later task is reached no later and with no more energy. All tasks that are on time in (i) remain on time, and the flown distance, hence the energy term γE, does not grow. The objective therefore loses the values v_j of the skipped tasks and nothing else, which is at most $\Lambda(\tau)$ and proves (ii). Part (iii) holds because the start shift does not decrease with τ and every tolerance is non-negative in a feasible plan.

4.5. Execution-aware delay cost

Under uncertain execution, task j is lost when $\sigma_j(\tau) + e_j > \mathrm{tol}_j$, where e_j is its finish-time error. The execution-aware cost replaces the indicator in Λ by the probability of this event. The error of task j is

modelled as $e_j \sim \mathcal{N}(m(h_j), s(h_j)^2)$, where $h_j = f_j - t_0$ is its lead time, the nominal finish time minus the time of the planning snapshot. The mean and the spread are linear in the lead time,

$$m(h) = b_0 + b_1 h, \qquad s(h) = s_0 + s_1 h, \tag{4}$$

because errors accumulate along a route. The expected dissemination latency $\bar{L}$ is added to the start shift. The expected loss and the execution-aware delay cost are

$$\Lambda_{\mathrm{R}}(\tau) = \sum_j v_j \, \Phi\!\left(\frac{\sigma_j(\tau) + \bar{L} + m(h_j) - \mathrm{tol}_j}{s(h_j)}\right), \qquad C_{\mathrm{R}}(\tau) = \Lambda_{\mathrm{R}}(\tau + H) - \Lambda_{\mathrm{R}}(\tau), \tag{5}$$

where Φ is the standard normal distribution function. Search and predictor are unchanged, and C_{R} replaces C_{delay} in Eq. (3).

Proposition 2. *(i) With $b_0 = b_1 = s_0 = s_1 = 0$ and $\bar{L} = 0$, $\Lambda_{\mathrm{R}} = \Lambda$, and the stopping rule takes exactly the decisions of the deterministic one. (ii) $\Lambda_{\mathrm{R}}(\tau)$ does not decrease with τ, so $C_{\mathrm{R}} \geq 0$. (iii) Under the error model, $\Lambda_{\mathrm{R}}(\tau)$ is the expected value of the tasks whose deadline is missed if every UAV follows its route.*

Proof. With zero spread, Φ is read as the indicator of a positive argument, which gives (i). The start shift does not decrease with τ and Φ is non-decreasing, which gives (ii). Part (iii) follows from the proof of Proposition 1, because task j misses its deadline exactly when its delay at the finish, $\sigma_j + e_j$ minus the absorbed waiting time, is positive, and the expectation of the indicator of this event is the probability in Eq. (5).

The parameters of Eq. (4) are fitted on development missions. The mean is estimated by least squares, and the spread by least squares on the absolute residuals scaled by $\sqrt{\pi/2}$, which is less sensitive to the heavy tail of the errors than squared residuals. The latency $\bar{L}$ is the mean of the latency distribution in tier B and the measured mean plan-acceptance latency in tier C.

4.6. Chance-constrained planning model

The delay cost prices the risk of a late plan, but the plan itself is still built for nominal execution. The fitted error models show that the nominal model is optimistic, since the mean error grows with the lead time (Table 4). The proposed replanner therefore plans for the calibrated execution. A planned finish time f_j is accepted only if

$$f_j + m(f_j - t_0) + z\, s(f_j - t_0) \;\leq\; d_j, \tag{6}$$

where $z \geq 0$ is a safety factor. The left-hand side depends on the finish time itself, so the buffer is small for tasks served soon and large for tasks served late in a route. Because m and s are linear, the constraint reduces to an effective deadline,

$$f_j \leq \tilde{d}_j, \qquad \tilde{d}_j = t_0 + \frac{d_j - t_0 - b_0 - z s_0}{1 + b_1 + z s_1}. \tag{7}$$

The planner, the repair and the search use $\tilde{d}_j$ instead of d_j, while the executor and the reward still use the true deadline. No other part of the planner changes.

Proposition 3. *Let $1 + b_1 + z s_1 > 0$. (i) A finish time satisfies Eq. (6) if and only if $f_j \leq \tilde{d}_j$. (ii) Under the error model of Eq. (4), a task planned to finish at $f_j \leq \tilde{d}_j$ meets its deadline with probability at least $\Phi(z)$ if its route starts as planned. (iii) With $b_0 = b_1 = s_0 = s_1 = 0$, $\tilde{d}_j = d_j$ and the planning model is the nominal one.*

Table 2: Scenario generator parameters (distances in m, times in s, energy in metres of flight).

Map side / horizon	10 000 / 1 800	Service time	$U[30, 90]$
UAV speed	$U[10, 16]$ m/s	Time-window slack	$U[180, 600]$
Energy capacity	$U[18, 28]$ km	Release delay	$U[0, 120]$
Energy reserve	10% of capacity	Rewards 1, 2, 3, 5	probabilities .4, .3, .2, .1
Hover cost	6 per second	Hotspots	3, std. 600, 50% of tasks
Tasks known at start	50%	Scales (UAVs/tasks)	5/25, 10/50, 20/140, 40/280

PROOF. Substituting Eq. (4) into Eq. (6) gives $f_j(1 + b_1 + zs_1) \le d_j - b_0 - zs_0 + (b_1 + zs_1)t_0$, and dividing by the positive factor gives (i). For (ii), the actual finish is $F_j = f_j + e_j$ with $e_j \sim \mathcal{N}(m(h_j), s(h_j)^2)$ and $h_j = f_j - t_0$. By (i), $d_j - f_j - m(h_j) \ge z\, s(h_j)$, hence $\Pr[F_j \le d_j] = \Phi\big((d_j - f_j - m(h_j))/s(h_j)\big) \ge \Phi(z)$. Part (iii) is immediate.

The effective deadline turns a chance constraint on every task into a deterministic constraint of the same form as the original one, so the repair, the insertion heuristics and the feasibility checks of the search are reused unchanged and the planning time does not grow. With $z = 0$, the model only corrects the mean error and plans for the median finish time. Larger values of z protect more tasks against lateness but leave fewer tasks feasible, so z trades the risk of a late service against the reward of attempting a task. It is chosen per calibrated disturbance level on development missions. Because the delay cost of Eq. (5) is evaluated on the effective deadlines, the mean error is already part of the tolerance, and the stopping rule uses the spread only, with $m \equiv 0$.

4.7. Replanning trigger

Errors accumulate between replanning calls. The proposed replanner therefore replans on every event and in addition at least every $T_{\max} = 60$ s of mission time, which corresponds to a rolling-horizon policy in the terminology of rescheduling [35, 36]. A deviation trigger, which replans when a finish time deviates from the plan by more than a threshold, was evaluated on development missions and gave the same utility, and is reported as an alternative.

5. Evaluation framework

5.1. Scenarios

Missions are created by a seeded generator with the parameters in Table 2. The map is a square with a side of 10 km and the depot at its centre, and half of the tasks are clustered around three hotspots. Half of the tasks are known at the start, and the others arrive in batches. Three fleet scales with 5, 10 and 20 UAVs and 25, 50 and 140 tasks are combined with three event profiles. The arrivals profile contains six arrival batches, the failures profile three batches and two UAV failures, and the deadlines profile three batches and four tightened deadlines. A fourth scale with 40 UAVs and 280 tasks is used for scalability only.

5.2. Static snapshots with exact execution

The stopping rules are first compared without execution uncertainty. Each mission is simulated once with a deterministic reference planner, and every replanning call after the start is stored with its previous plan. All methods are evaluated on these snapshots, and regret is measured against the best known objective of each snapshot, which includes a 3 s ALNS run and, for the small and medium scales, the OR-Tools routing solver [53] with guided local search [54]. Because all ALNS variants follow the same search trajectory, whole families of each rule are evaluated exactly offline, with 12 fixed budgets, 10 stagnation patience values and 9 compute prices, and each baseline is compared with AnytimeSwarm at the same mean computation. This benchmark uses its own pre-registered test seeds 2000–2029 (1562 snapshots of 270 missions), and it was reproduced on the final software platform before the execution-aware study (Section 6.2).

Table 3: Disturbance and latency levels of tier B.

Execution	accel. (m/s^2)	speed c.v.	wind (m/s)	service c.v.	Latency	median (s)	$\sigma_{\log}$	loss / timeout
nominal	∞	0	0	0	none	0	–	–
low	3.0	0.03	2	0.05	low	0.2	0.5	–
mid	2.0	0.07	5	0.15	high	2.0	0.8	5% / 5 s
high	1.5	0.12	8	0.25				

5.3. Tier B: kinematic execution with disturbances and latency

Tier B executes each published plan in continuous mission time. The duration of a leg follows a trapezoidal speed profile with a horizontal acceleration limit. The cruise airspeed of each (UAV, task) leg deviates from the nominal speed by a multiplicative error, and a constant wind of fixed magnitude and random direction changes the ground speed through the wind triangle. Service durations are multiplied by a random factor. Energy is counted as metres flown through the air plus the hover cost. The planner never sees these disturbances. It receives the state of each UAV as telemetry, and the remaining time of a committed task is estimated with the nominal model from the actual position of the UAV. The dissemination latency of a plan is lognormal, and each message is lost with a given probability and resent after a timeout. Table 3 lists the disturbance and latency levels. They span a range from mild to severe conditions and are not claimed to represent a particular vehicle.

All random draws use common random numbers [13]. The speed error of a leg depends only on the mission seed, the UAV and the task, the service factor only on the mission seed and the task, the wind only on the mission seed, and the latency only on the mission seed and the event. Two planners that assign the same task to the same UAV therefore experience the same disturbance, independently of the number of replanning calls. With zero disturbance, zero latency and the event trigger, tier B reproduces the discrete-event simulator of the original formulation exactly, which is verified by unit tests.

5.4. Tier C: PX4 software-in-the-loop with Gazebo and ROS 2

Tier C flies missions with three X500 quadrotors, each controlled by its own PX4 v1.17 autopilot in software-in-the-loop mode [14], in Gazebo Harmonic [15], on ROS 2 Jazzy [16] (Fig. 2). A uXRCE-DDS bridge exposes the internal topics of each autopilot to ROS 2. Three nodes form the mission stack. The mission manager runs the scenario timeline, builds the planning snapshot from the fleet state, calls the planner and releases the plan. The executor streams offboard setpoints to every autopilot at 10 Hz, so that PX4 does not leave offboard mode while the planner computes. It flies each leg with a carrot that moves along the straight leg at the commanded ground speed, holds the UAV at the task for the service duration and reports service start, finish and skips. A visualisation node draws tasks, routes and planning decisions in RViz 2 and as coloured pillars in the Gazebo scene. Planning time is measured with a monotonic wall clock, and every mission time is simulation time.

A tier C mission is a tier B scenario with three UAVs and fifteen tasks, which keeps the ratio of five tasks per UAV of the small scale, rescaled by $S_X = 0.1$ in length and $S_T = 0.25$ in time. Speeds are multiplied by S_X/S_T, energy by S_X, the hover cost by S_X/S_T and the distance price by $1/S_X$. Every dimensionless ratio of the decision problem is preserved, so a tier C mission is the same decision problem as its tier B counterpart, flown in a 1 km world in 450 s with cruise speeds of 4–6.4 m/s. A planning delay of τ in tier C corresponds to τ/S_T in tier B, so the latency factors are related by $\kappa_C = S_T \kappa_B$. Disturbances use the same common random numbers as tier B. The commanded ground speed of each leg follows the speed error and the wind triangle, with the wind scaled by S_X/S_T, and the service duration follows the same factor. Acceleration, attitude dynamics, position tracking and communication are those of the simulated vehicles. Gazebo runs in lockstep with PX4 at twice real time, which requires $\kappa_C \geq 2$ so that the emulated planning delay dominates the real one.

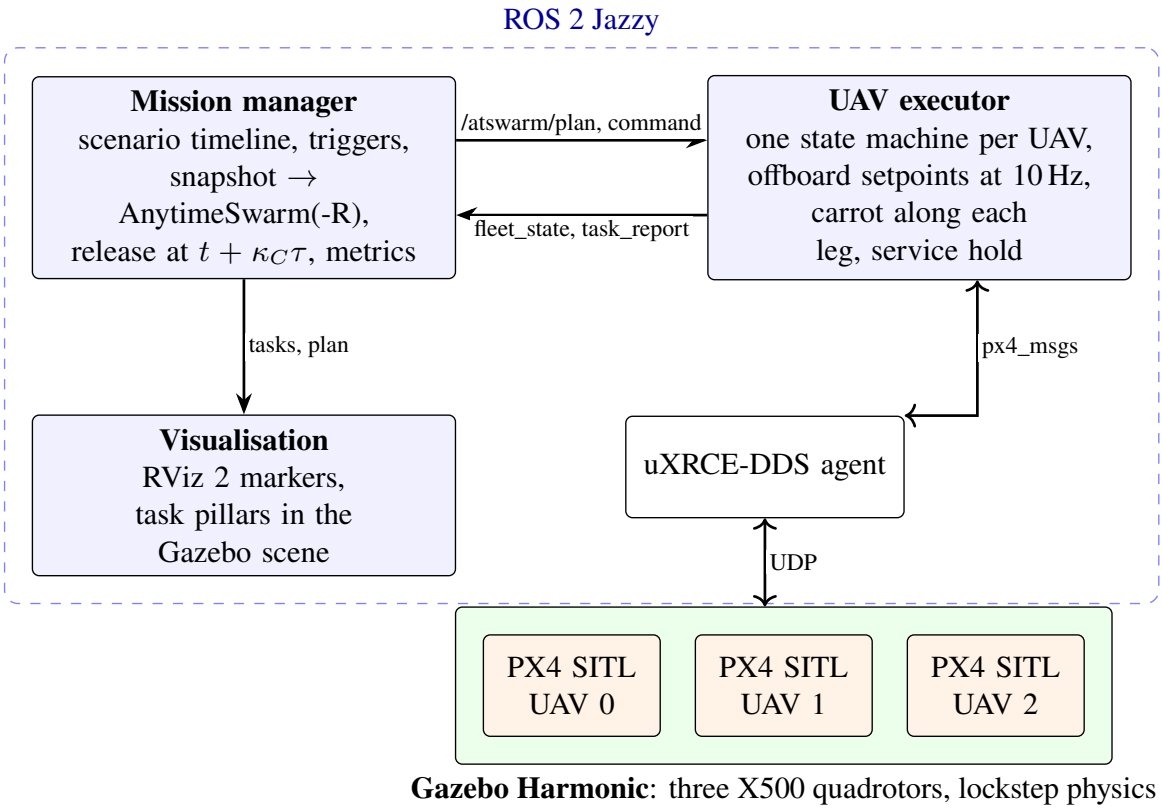


Figure 2: Tier C architecture. The planner and the mission timeline run in the mission manager; the executor streams offboard setpoints to three PX4 autopilots that fly X500 quadrotors in Gazebo.

5.5. *Compared methods*

All methods share the feasibility-first repair, the ALNS operators and parameters, the initial plan and the random seed, and every call has a hard budget of 250 ms. They differ in the stopping rule, the planning model and the trigger.

Stopping rules. Regret-2 repair without search is the constructive floor. ALNS is run with a fixed budget of 250 ms [4], with a fixed budget of 40 ms, which equals the mean computation of AnytimeSwarm on development data, with a budget proportional to the number of open tasks (1 ms per task, chosen on development missions among 0.5, 1 and 2 ms), and with a stagnation rule that stops after 20% of the budget without improvement. Two stopping rules from the metareasoning literature are re-implemented with the same time cost as the proposed rule, that is, the delay cost and the price of computation, so that only the prediction of the gain differs. The performance profile rule [8, 17] predicts the gain from a static profile of the expected improvement over elapsed time and number of open tasks, estimated on 606 development planning calls. The online prediction rule follows Svegliato et al. [10] and fits the curve $q(t) = a - b\, e^{-ct}$ to the best objective found so far at every check, by least squares over a logarithmic grid of rates. AnytimeSwarm uses the learned predictor with the deterministic delay cost and AnytimeSwarm-R with the execution-aware cost.

Planning models and triggers. The nominal planning model uses the deadlines as given. The constant-buffer model subtracts 60 s from every deadline, chosen on development missions among 30, 60 and 120 s, as is common for robust schedules with time buffers [12]. The proposed chance-constrained model uses Eq. (7)

with $z = 0.5$ under mid and $z = 0$ under high disturbances, selected on development missions. Triggers are event-only or event plus periodic replanning every 60 s.

5.6. Two pre-registered studies

The evaluation consists of two pre-registered held-out studies, each with its own seeds, written protocol and frozen code. Development used seeds 0–9 in tier B and 0–2 in tier C. The predictor was trained on development trajectories, and the compute price $\mu = 10^{-5}$ was selected by a rule stated before any test run. The tier B error models were fitted on 90 development missions per level flown by ALNS with 40 ms, and the tier C models on nine development flights per level (Table 4). Before each study, the experiment configurations, the models and the experiment code were frozen with SHA-256 hashes, and the held-out runs checked these hashes before starting.

Study 1 asks whether the stopping rule alone can be made execution-aware. It compares AnytimeSwarm-R with the stopping rules of the first group on the nominal planning model, on held-out seeds 3000–3029 in tier B (270 missions over three scales and three profiles, and 90 missions with 40 UAVs) and 4000–4009 in tier C (30 missions per method). Its primary matrix crosses two execution levels (low, mid), two latency levels (none, high), two latency factors ($\kappa_B = 10, 50$) and two triggers (event, periodic every 60 s). Its primary hypothesis is that AnytimeSwarm-R serves more reward than ALNS with 40 ms, AnytimeSwarm and the stagnation rule.

Study 2 evaluates the proposed replanner. It was designed after study 1, on development missions only, and uses new held-out seeds 5000–5029 in tier B and 6000–6009 in tier C. Its primary matrix crosses two execution levels (mid, high), high latency, three latency factors ($\kappa_B = 10, 50, 200$) and two time-window factors (original windows and windows halved), which gives twelve configurations and 270 missions each. Fifteen methods are compared: the proposed replanner, the ten baselines of Section 5.5 with the periodic trigger, the conventional planner that uses ALNS with 40 ms, the nominal model and the event trigger, and three ablations. The primary hypothesis H1 is that the proposed replanner serves more reward than each of six baselines (ALNS with 40 ms, the size-scaled budget, online prediction, the performance profile, AnytimeSwarm-R and the constant buffer, all with the periodic trigger). H2 compares it with the conventional planner. H3 tests whether the proposed stopping rule is equivalent to a fixed budget of 40 ms on the proposed planning model, with a margin of ±0.5 percentage points. In tier C, H4 compares the proposed replanner with ALNS with 40 ms and with the conventional planner on 30 missions.

In both studies, the utility difference of a pair of methods is averaged per mission over the configurations of the matrix, and the mean over missions is tested with a paired bootstrap [55] of 10 000 resamples. Families of primary comparisons are corrected with the Holm procedure [56], and equivalence is tested with two one-sided tests, that is, a 90% interval inside the margin. All other comparisons are descriptive. A tier C run in which a vehicle does not reach its altitude within 90 s of simulation time, or whose real-time factor leaves $[1.6, 2.2]$, is repeated once with the same seed. The protocols are available with the code, together with a list of the deviations from them. Tier B runs four missions in parallel on an Intel Core i7-11800H laptop, with all methods of a configuration run in turn, so that they share the same machine load. All runs use Python 3.12 and pinned package versions. Every plan is checked for feasibility before publication, and a separately written validator that shares no code with the planner found no infeasible plan among 4 779 development plans of all methods.

6. Results

6.1. Exact execution: stopping rules at equal computation

On the 1562 test snapshots, regret-2 repair finished within about one millisecond but left a regret of 0.98%. With a budget of 250 ms, ALNS with the full budget reached 0.06%, whereas AnytimeSwarm reached 0.08% with 16% of the budget, that is, with 84% less computation. Because the stopping rules use different amounts

Table 4: Finish-time error models fitted on development missions: $m(h) = b_0 + b_1 h$, $s(h) = s_0 + s_1 h$ (seconds of the respective tier; n executed tasks).

Tier	Level	n	b_0	b_1	s_0	s_1
B	low	5399	8.91	−0.0044	0.00	0.0904
B	mid	4991	16.84	0.0354	1.58	0.1951
B	high	4253	20.79	0.2923	0.00	0.6389
C	low	90	−1.23	0.0281	1.30	0.0545
C	mid	89	2.36	0.0297	0.00	0.2303

Table 5: Regret at equal computation per fleet scale (UAVs/tasks). Baselines run for the mean time of AnytimeSwarm on the same scale; †: 95% interval of the paired difference below zero. Right: computation a baseline needs to match AnytimeSwarm.

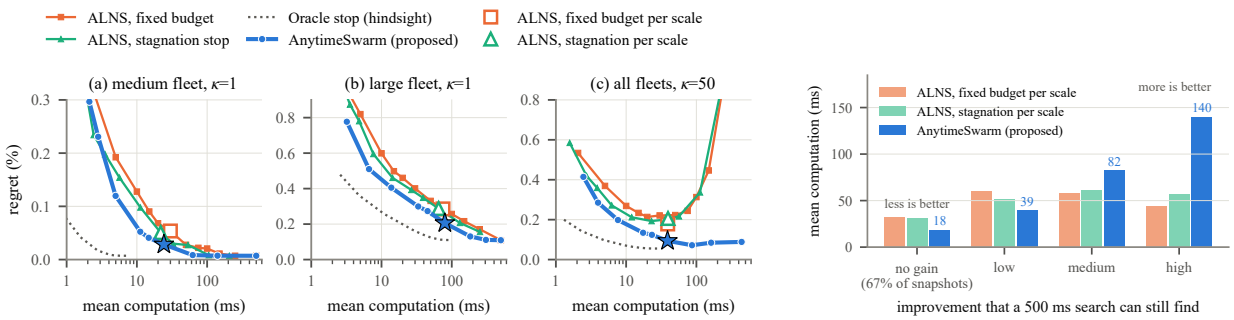
		Regret at equal computation (%)			Computation to match (ms)	
Scale	Time (ms)	Proposed	Fixed	Stagnation	Fixed	Stagnation
Small (5/25)	18	0.006	0.030†	0.029†	50 (2.7×)	51 (2.8×)
Medium (10/50)	25	0.030	0.062†	0.045†	49 (2.0×)	39 (1.5×)
Large (20/140)	81	0.213	0.290†	0.274†	172 (2.1×)	135 (1.7×)
40 UAVs (40/280)	151	0.205	0.256†	0.234†	299 (2.0×)	258 (1.7×)
Small to large	42	0.083	0.141†	0.127†	143 (3.4×)	99 (2.4×)

of computation, they are best compared at equal computation. Table 5 makes this comparison for each fleet scale, with the fixed budget and the stagnation rule evaluated at the mean computation that AnytimeSwarm used on the same scale. AnytimeSwarm reached a lower regret on every scale, and all paired intervals lie below zero. The reduction was 33% to 52% on the medium scale, 22% to 27% on the large scale and 12% to 20% with 40 UAVs, a scale that was never used for training. Read in the other direction, the two baselines needed between 1.5 and 2.8 times the computation of AnytimeSwarm to reach its regret. Over all scales, AnytimeSwarm reached 0.083% against 0.112% to 0.130% for four baselines that include a fixed budget and a stagnation patience tuned per fleet scale on development data.

Fig. 3 shows where this advantage comes from. On the medium and large scales, the quality–compute frontier of AnytimeSwarm lies below those of fixed budgets and stagnation stopping over the range of practical computation (Fig. 3a). Grouped by the improvement that a 500 ms search can still find (Fig. 3b), AnytimeSwarm spent 18.2 ms on snapshots that cannot be improved and 140.0 ms on those with the largest available improvement. Only 29% of its computation went to snapshots without any possible gain, compared with more than half for the baselines. The learned predictor is essential for this allocation. It reached an area under the ROC curve [57] of 0.926 for the event that the next 10 ms bring an improvement, compared with 0.757 for an extrapolation of the recent improvement rate.

6.2. Reproduction on the evaluation platform

The exact-execution benchmark was produced on Windows. Before the execution-aware study, its development pipeline was re-run on the platform of all later experiments (Ubuntu 24.04 under WSL2, identical Python and package versions). The deterministic part was reproduced bit for bit, except for differences of one unit in the last place of some energy values, which stem from the floating-point libraries of the two operating systems. The search ran 21% more iterations per millisecond on the new platform. The improvement predictor, re-trained on the new timing traces, reached the same compute price by the pre-registered rule, and the conclusions of the equal-computation comparison agreed in sign and significance in 24 of 24 comparisons. The frozen predictor was used unchanged in all later experiments.



(a) (b)

Figure 3: Exact execution, test seeds 2000–2029: (a) quality–compute frontiers on identical search trajectories (open markers: baselines tuned per scale; star: selected AnytimeSwarm setting); (b) mean computation per snapshot, grouped by the improvement that a 500 ms search can still find.

6.3. Study 1: execution-aware stopping on the nominal planning model

6.3.1. Tier B primary matrix

The primary matrix comprises 34 560 mission runs, that is, 270 held-out missions in 16 configurations with eight methods. Table 6 lists utility, lost tasks and computation per method, and Fig. 4 relates utility to computation and shows the difference in every configuration.

Search pays off clearly under uncertain execution. Regret-2 repair reached a utility of 68.86%, and AnytimeSwarm-R reached 72.25%, which is +3.39 percentage points (pp) or 4.9% more. The full budget of 250 ms was the weakest search method, with 71.48%, and AnytimeSwarm-R served +0.77 [+0.31, +1.25] pp more reward than it. The loss of the long budget stems from late plans. A task counts as lost to delay when it is part of the computed plan but can no longer be served on time when the plan is released. With $\kappa_B = 50$, the full budget lost 5.36 tasks per mission in this way, against 1.28 for AnytimeSwarm-R, that is, 76% fewer. Without its delay cost, AnytimeSwarm lost 3.50 tasks per mission in this way, but its utility was not lower (72.13% against 72.04%). The delay cost thus prevents losses from late plans without a measurable gain in utility on this matrix.

Among the remaining search methods, the differences are small. All seven search methods lie within 0.77 pp of each other. AnytimeSwarm-R reached the highest mean utility with the least computation of the adaptive rules, 41.1 ms per call against 48.2 ms for AnytimeSwarm and 59.9 ms for the stagnation rule. The reduction of 15% relative to AnytimeSwarm is significant (−7.1 [−7.7, −6.5] ms per call), so the second hypothesis holds. The primary hypothesis does not hold at the pre-registered level. The pooled advantage of AnytimeSwarm-R was +0.32 [+0.01, +0.63] pp over ALNS with 40 ms, +0.21 [−0.08, +0.52] pp over AnytimeSwarm and +0.28 [−0.11, +0.67] pp over the stagnation rule, with Holm-adjusted p-values of 0.128, 0.308 and 0.308. Only the first interval excludes zero before correction. Per configuration and without correction for multiple testing, AnytimeSwarm-R was better than ALNS with 40 ms in 2 of the 16 configurations, better than the stagnation rule in 1 and never different from AnytimeSwarm, which is consistent with chance (Fig. 4b). The ablation that keeps only the latency term reached 72.02%, close to AnytimeSwarm. Together with the misspecification runs below, this suggests that a non-zero error spread mainly makes the rule stop earlier, rather than that the calibration of the spread matters.

The fixed budget of 40 ms is a strong reference, because it was matched to the mean computation of AnytimeSwarm on development data. Its weakness lies in the allocation across fleet scales. AnytimeSwarm-R used 24 ms per call on the small scale and 75 ms on the large scale. On the large scale, it served +0.59 [+0.34, +0.82] pp more reward than the budget of 40 ms and was on a par with the full budget (−0.06 [−0.38, +0.26] pp). On the small scale, it served +1.79 [+0.65, +3.00] pp more than the full budget and was on a par with the budget of 40 ms (+0.33 [−0.40, +1.09] pp). A single fixed budget is therefore either too long for

Table 6: Tier B primary matrix: 270 held-out missions × 16 configurations. Utility and lost tasks per mission, mean planning time per call, and pooled paired difference of AnytimeSwarm-R minus each method (percentage points, 95% bootstrap interval; Holm-adjusted p for the three primary comparisons).

Method	Utility (%)	Late	Skipped	Calls	ms/call	AnytimeSwarm-R minus method (pp)
Regret-2 repair	68.86	6.64	18.35	15.6	1.0	+3.39 [+2.76, +4.04]
ALNS, 250 ms	71.48	6.15	20.58	15.5	250.1	+0.77 [+0.31, +1.25]
ALNS, 40 ms	71.93	6.65	20.97	15.5	40.1	+0.32 [+0.01, +0.63], p_{Holm} 0.128
ALNS, stagnation	71.97	6.48	20.92	15.5	59.9	+0.28 [−0.11, +0.67], p_{Holm} 0.308
AnytimeSwarm w/o delay cost	72.13	6.35	20.49	15.5	54.1	+0.12 [−0.20, +0.45]
AnytimeSwarm	72.04	6.64	21.32	15.5	48.2	+0.21 [−0.08, +0.52], p_{Holm} 0.308
AnytimeSwarm-R, latency only	72.02	6.65	21.31	15.5	48.2	+0.23 [−0.06, +0.54]
AnytimeSwarm-R (proposed)	72.25	6.56	21.15	15.5	41.1	–

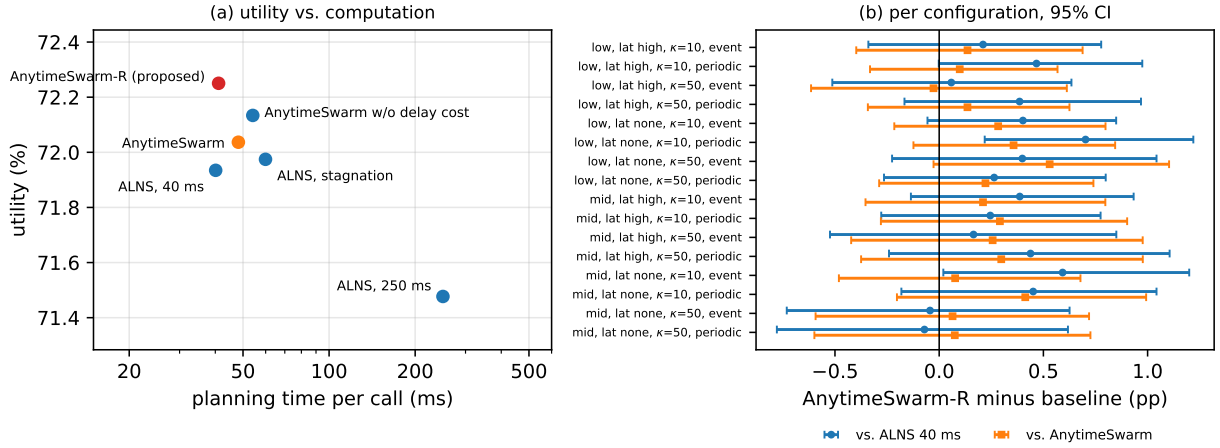


Figure 4: Tier B primary matrix, held-out seeds 3000–3029: (a) mean utility against mean planning time per call over all 16 configurations (regret-2 repair, at 68.86% and 1 ms, is not shown); (b) paired difference of AnytimeSwarm-R minus ALNS with 40 ms and minus AnytimeSwarm per configuration (execution level, latency level, κ_B, trigger) with 95% bootstrap intervals over 270 missions.

small fleets or too short for large ones, whereas the adaptive rules find a suitable amount of computation on each scale without tuning. The advantage over ALNS with 40 ms was +0.43 [+0.11, +0.76] pp for $\kappa_B = 10$ and +0.20 [−0.19, +0.61] pp for $\kappa_B = 50$, and +0.36 [+0.01, +0.73] pp under low and +0.27 [−0.13, +0.66] pp under mid disturbances.

6.3.2. Replanning triggers

The trigger mattered more than the stopping rule. Replanning every 60 s in addition to the events raised the number of planning calls from 6.8 to 24.2 per mission and the utility of every method, by +1.29 pp to +1.56 pp, with all intervals above zero (+1.56 [+1.26, +1.84] pp for AnytimeSwarm-R, +1.47 [+1.19, +1.75] pp for ALNS with 40 ms). The third hypothesis therefore holds. Periodic replanning probably helps because it corrects the accumulated deviations of the executed routes, which the event trigger only notices at the next event. The advantage of AnytimeSwarm-R over ALNS with 40 ms was +0.27 [−0.07, +0.62] pp with the event trigger and +0.36 [+0.03, +0.70] pp with the periodic trigger. The two values are close, and the protocol did not test their difference.

6.3.3. Scalability to 40 UAVs

With 40 UAVs and 280 tasks, 90 held-out missions were run under mid disturbances in 4 configurations of latency level and latency factor. All five methods reached between 68.82% and 69.23%. AnytimeSwarm-R reached 69.16% with 127 ms per call. Its pooled difference was +0.02 [−0.26, +0.30] pp to ALNS with 180 ms, +0.34 [+0.05, +0.62] pp to AnytimeSwarm and −0.07 [−0.30, +0.16] pp to the stagnation rule. After Holm correction over the four comparisons of the pre-registered analysis, none of them is significant ($p_{\text{Holm}} = 0.086$ for AnytimeSwarm), so the fifth hypothesis is not supported either. AnytimeSwarm-R again needed less computation than AnytimeSwarm (−9.0 [−11.7, −6.2] ms per call). The stagnation rule stopped earliest, after 100 ms, and was significantly better than AnytimeSwarm-R in 1 of the 4 configurations, which had no latency and $\kappa_B = 50$. On this scale, the rules used between 100 and 250 ms per call, and utility hardly changed within this range of computation. The equal-computation advantage that AnytimeSwarm showed with 40 UAVs under exact execution (Table 5) therefore did not translate into a utility gain under uncertain execution.

6.3.4. Sensitivity analyses

The sensitivity runs use the held-out missions of the primary matrix without latency and are descriptive. They vary the execution level beyond the primary matrix, the latency factor and the trigger in 8 configurations, and the error model in 4 further configurations.

Under high disturbances, which lie outside the range of the primary matrix but within that of the fitted error models, utility fell to about 48.93% for all methods. The combination of high disturbances and a slow planner ($\kappa_B = 50$) gave the largest advantage of the risk-aware cost in this study. AnytimeSwarm-R served +0.96 [+0.29, +1.66] pp more reward than AnytimeSwarm and used 49 ms instead of 58 ms per call. Its difference to ALNS with 40 ms was +0.14 [−0.53, +0.84] pp and to the stagnation rule +0.43 [−0.33, +1.21] pp, so the gain over AnytimeSwarm brought it to the level of the baselines rather than beyond them. With $\kappa_B = 1$, planning time hardly delays the plan, both delay costs are close to zero because they scale with the shift $\kappa\tau$, and the two rules behave alike. AnytimeSwarm-R then differed from AnytimeSwarm by +0.01 [−0.13, +0.15] pp and −0.5 ms per call.

Replanning when a UAV deviated by more than 30 s from its planned finish time raised the number of calls from 6.8 to 13.2 per mission and the utility by +0.86 [+0.58, +1.14] pp for AnytimeSwarm-R and +0.86 [+0.54, +1.18] pp for ALNS with 40 ms. Like the periodic trigger, it helped both stopping rules by a similar amount. Under the deviation trigger, the difference of AnytimeSwarm-R was +0.10 [−0.30, +0.52] pp to ALNS with 40 ms and +0.29 [−0.05, +0.65] pp to AnytimeSwarm.

The utility of AnytimeSwarm-R was insensitive to the error model. With the error model of the low, mid or high level under low and mid disturbances, its difference to AnytimeSwarm ranged between +0.07 [−0.31, +0.47] pp and +0.18 [−0.18, +0.55] pp, the correctly specified model was not better than a wrong one, and the mean utilities of all versions lay within 0.11 pp. Setting the error model to zero gave −0.13 [−0.32, +0.03] pp. The four misspecification configurations also belong to the primary matrix, so the correctly specified method was run twice on the same missions. The two runs differ only in the measured planning times, because the stopping rule reads a wall clock. They gave identical utility in 73% of the missions, and the mean utility of a method differed by up to 0.12 pp between the runs. Differences of this size between stopping rules should therefore not be read as systematic.

6.3.5. Tier C: PX4 software-in-the-loop with Gazebo

All 120 held-out flights of tier C were completed, that is, 30 missions with four methods each. Two runs did not reach their altitude within 90 s and were repeated as specified. A further 4 runs had a real-time factor between 1.43 and 1.52, below the validity range of 1.6 to 2.2 stated in the protocol. The protocol did not specify how to treat such runs, so they were repeated once with the same seed, like a technical failure, and the original runs were kept. The repeated runs reached the same utility as the original ones, and the comparisons

Table 7: Tier C (PX4 SITL, Gazebo, ROS 2): 30 held-out missions per method, mid disturbances, $\kappa_B = 50$. Utility, tasks lost to late plans and skipped at departure per mission, planning time per call; tier B utility on the same rescaled missions for comparison.

Method	Utility C (%)	Lost to delay	Skipped	ms/call	Utility B (%)
ALNS, 250 ms	51.88	0.43	0.57	250.1	50.17
ALNS, 40 ms	54.69	0.03	0.50	40.1	51.72
AnytimeSwarm	53.70	0.03	0.43	47.4	51.90
AnytimeSwarm-R (proposed)	54.18	0.00	0.37	34.7	52.55

below are identical with either set (−0.51 [−3.24, +2.18] pp and +0.48 [−2.72, +3.46] pp with the original runs). The real-time factor of the analysed runs was 1.95 on average and at least 1.67. A released plan reached all vehicles after 0.152 s of simulation time on average, which is included in the expected latency of AnytimeSwarm-R.

Table 7 summarises the results. AnytimeSwarm-R reached a utility of 54.18%, against 54.69% for ALNS with 40 ms, 53.70% for AnytimeSwarm and 51.88% for the full budget. The paired differences were −0.51 [−3.24, +2.18] pp to ALNS with 40 ms and +0.48 [−2.72, +3.46] pp to AnytimeSwarm, with Holm-adjusted p-values of 1.000 and 1.000. The fourth hypothesis is therefore not supported. With 30 missions, the intervals are much wider than in tier B and cannot resolve differences of the size seen there. Two effects of tier B reappear in tier C. The full budget again had the lowest mean utility, although its interval includes zero (+2.30 [−1.32, +5.71] pp to AnytimeSwarm-R), and it was the only method that lost a noticeable number of tasks to late plans, 0.43 per mission against at most 0.03 for the others. AnytimeSwarm-R again computed least among the adaptive rules, with 34.7 ms per call against 47.4 ms for AnytimeSwarm, a reduction of 27% (−12.7 [−18.8, −7.2] ms).

The same 30 missions were also run in tier B with the rescaled parameters of tier C. Tier B predicted the outcome of a flight well. The utility of a run in tier C correlated with that of the same run in tier B with $r = 0.84$ (120 runs), and the mean utility per mission with $r = 0.89$ (Fig. 5b). Tier C served slightly more reward on average, 53.61% against 51.58%, so the kinematic model of tier B was, if anything, slightly pessimistic for these missions. The small differences between the methods were not predicted as well. The method means of the two tiers had a Kendall rank correlation of 0.33, and the sign of the per-mission difference between AnytimeSwarm-R and ALNS with 40 ms agreed in 40% of the missions, counting ties as agreement only when both differences are zero. Part of the per-run correlation is by construction, because both tiers draw the wind, speed and service disturbances from the same random numbers, and much of it reflects the different difficulty of the missions. The correlation was first computed with an erroneous pairing of runs in the analysis script, which was corrected after the results had been seen; the hypothesis tests were not affected. In tier B, the advantage of AnytimeSwarm-R on these missions was +0.82 [−0.27, +2.16] pp over ALNS with 40 ms and +0.65 [+0.00, +1.50] pp over AnytimeSwarm (Fig. 5a). Both tiers thus agree that the methods differ by at most a few percentage points on this scale, and that the full budget is the weakest search method, while the order of the three short-computation methods is within the noise of both tiers.

6.4. Study 2: the proposed replanner against baselines from the literature

6.4.1. Tier B primary matrix

Study 2 comprises 48 600 tier B mission runs, that is, 270 held-out missions in 12 configurations with 15 methods. Table 8 lists utility, late and lost tasks and computation per method. Fig. 6 shows the paired difference of the proposed replanner minus every other method and relates utility to computation.

The proposed replanner reached the highest utility, 52.15%. Its advantage over the six baselines of the primary hypothesis ranged from +1.35 pp over the constant buffer to +2.87 pp over ALNS with 40 ms, and every Holm-adjusted p-value was below 0.001. The primary hypothesis H1 therefore holds for 6 of the six comparisons. In relative terms, the proposed replanner served 5.3% more reward than the performance profile

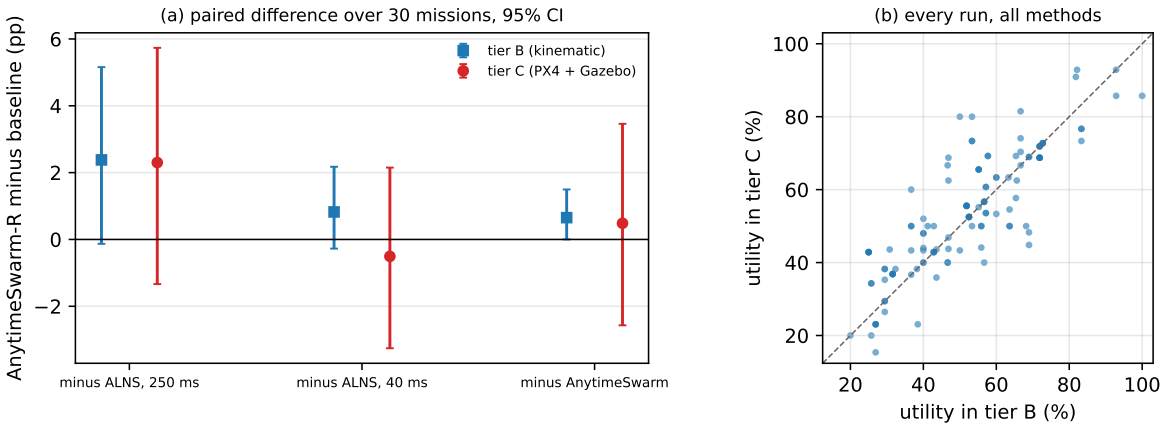


Figure 5: Tier B against tier C on the same 30 held-out missions (three UAVs, mid disturbances, $\kappa_B = 50$): (a) paired difference of AnytimeSwarm-R minus each baseline with 95% bootstrap intervals; (b) utility of every run in tier C against the same run in tier B (dashed: equality).

rule, 5.5% more than online performance prediction, 5.2% more than the size-scaled budget and 2.6% more than the constant buffer. Against the conventional planner, which combines ALNS with 40 ms, the nominal model and the event trigger, the advantage was +4.44 [+3.82, +5.05] pp, or 9.3%, so H2 holds as well.

The advantage comes from fewer late tasks. The proposed replanner served 5.72 tasks per mission after their deadline, against 11.22 for ALNS with 40 ms, that is, 49% fewer, and 6.53 for the constant buffer. It lost no task to a late plan, whereas ALNS with 40 ms lost 3.58 and the full budget of 250 ms lost 11.58 tasks per mission. On the nominal model, the stopping rules hardly differed. ALNS with 40 ms, the size-scaled budget, the performance profile, online prediction, AnytimeSwarm and AnytimeSwarm-R all lay between 49.28% and 49.66%, although their computation per call ranged from 11.2 ms to 58.1 ms (Fig. 6b). Correcting the deadlines was worth more than any choice of stopping rule.

6.4.2. Ablations

Three ablations remove one component of the proposed replanner at a time (Table 8, bottom). With a fixed budget of 40 ms instead of the proposed stopping rule, utility changed by +0.22 [+0.07, +0.37] pp. The 90% interval of this difference, +0.22 [+0.09, +0.35] pp, lies inside the pre-registered margin of ±0.5 pp, so the stopping rule is equivalent to the fixed budget on the chance-constrained model (H3), while it used +20.9 [+17.5, +24.3] ms more per call. The stopping rule still guards against long budgets. At $\kappa_B = 200$, the full budget of 250 ms lost 23.42 tasks per mission to late plans, and the advantage of the proposed replanner over it grew from +2.44 [+1.84, +3.03] pp at $\kappa_B = 10$ to +11.34 [+10.78, +11.91] pp at $\kappa_B = 200$.

With the event trigger instead of the periodic one, the proposed replanner made 6.8 instead of 23.3 planning calls per mission and lost +0.22 [+0.06, +0.38] pp. On the nominal model of study 1, the periodic trigger had added +1.29 pp to +1.56 pp. The chance-constrained model thus seems to absorb most of the deviations that periodic replanning corrects. With $z = 0$, that is, with the mean correction only, the difference was +0.18 [−0.06, +0.41] pp and not significant. Since $z = 0$ is also the frozen value under high disturbances, this ablation differs only under mid disturbances. There, the spread term helped with the original windows (+1.51 [+0.86, +2.16] pp) and hurt with halved windows (−0.82 [−1.52, −0.11] pp). Most of the advantage of the proposed replanner therefore comes from the planning model, namely from moving every deadline by the predicted finish-time error, and not from the stopping rule or the trigger.

Table 8: Tier B, held-out missions of study 2: 270 missions × 12 configurations per method. Utility, tasks served late (Late), tasks lost because a plan arrived too late (Lost), planning calls and computation per call; difference of the proposed method minus each method, pooled over configurations (percentage points, 95% paired bootstrap interval; Holm-adjusted p for the six primary comparisons, marked *).

Method	Utility (%)	Late	Lost	Calls	ms/call	Proposed minus method (pp)
Proposed	52.15	5.72	0.00	23.3	61.0	–
Conventional (40 ms, event-only)	47.71	10.31	3.32	6.8	40.2	+4.44 [+3.82, +5.05]
Regret-2 repair (no search)	47.95	11.19	2.34	23.2	0.5	+4.21 [+3.57, +4.84]
ALNS, fixed 250 ms [4]	46.23	9.78	11.58	22.9	250.1	+5.92 [+5.41, +6.44]
ALNS, fixed 40 ms [4]	49.28	11.22	3.58	23.3	40.1	+2.87 [+2.29, +3.45], p <0.001*
ALNS, size-scaled budget	49.58	11.18	3.75	23.3	17.7	+2.57 [+1.99, +3.17], p <0.001*
ALNS, stagnation stop	49.04	10.98	5.50	23.2	55.8	+3.11 [+2.54, +3.69]
Performance profile [8, 17]	49.54	11.24	3.52	23.2	30.2	+2.62 [+2.04, +3.19], p <0.001*
Online performance prediction [10]	49.45	11.17	2.75	23.2	11.2	+2.70 [+2.13, +3.28], p <0.001*
AnytimeSwarm	49.31	11.17	3.89	23.2	58.1	+2.84 [+2.29, +3.40]
AnytimeSwarm-R	49.66	11.21	3.73	23.2	50.9	+2.49 [+1.92, +3.06], p <0.001*
Constant 60 s buffer [12]	50.81	6.53	0.00	23.4	40.1	+1.35 [+0.93, +1.77], p <0.001*
Ablations of the proposed method						
Proposed, event-only trigger	51.93	5.85	0.00	6.8	57.8	+0.22 [+0.06, +0.38]
Proposed model, fixed 40 ms stop	51.93	5.74	0.00	23.4	40.1	+0.22 [+0.07, +0.37]
Proposed, mean correction only	51.97	6.84	0.00	23.3	58.8	+0.18 [−0.06, +0.41]

6.4.3. Regimes

Table 9 and Fig. 7 break the utility down by disturbance level, time-window factor and latency factor. Per configuration and without correction for multiple testing, the proposed replanner was better than ALNS with 40 ms in 8 of the 12 configurations and worse in 2, better than the constant buffer in 9 and worse in 3, and better than the conventional planner in 10 and worse in 0. The advantage over ALNS with 40 ms was +2.58 [+1.93, +3.22] pp, +2.71 [+2.10, +3.33] pp and +3.33 [+2.74, +3.91] pp for $\kappa_B = 10$, 50 and 200, and the advantage over online prediction stayed between +2.76 [+2.15, +3.36] pp and +2.57 [+1.97, +3.16] pp.

The advantage was largest with the original windows. Under high disturbances, the proposed replanner served +6.22 [+5.17, +7.29] pp more reward than ALNS with 40 ms and +3.03 [+2.01, +4.11] pp more than the constant buffer. The losses all occurred under high disturbances with halved windows. In this regime, the proposed replanner reached 33.38%, −2.01 [−2.80, −1.21] pp relative to the constant buffer and −1.57 [−2.52, −0.64] pp relative to ALNS with 40 ms. It served 24.3 tasks per mission, of which 6.2 late, against 25.3 and 8.4 for the constant buffer. The chance-constrained model was thus too cautious when the windows were short compared with the predicted finish-time error. The effective deadline of many tasks then probably falls before the earliest possible arrival, so the planner gives up tasks that a less cautious plan still serves in time. The value of z was selected per disturbance level on development missions, averaged over both window widths. Selecting it also per window width, or adapting it online, is a natural extension.

6.4.4. Tier C: PX4 software-in-the-loop with Gazebo

All 120 held-out flights of study 2 were completed, that is, 30 missions with four methods each. One run failed to start and was repeated. As specified in the protocol, 46 runs with a real-time factor outside 1.6 to 2.2 were repeated once with the same seed, and the original runs were kept. The laptop was also used for other work during the campaign, and many repeats ran slower than the original runs. Of the 120 analysed runs, 77 lie in the validity range. The real-time factor was 1.64 on average, with quartiles of 1.41–1.95 and a minimum of 0.83. Two sensitivity analyses that were not pre-registered therefore repeat the comparisons with the original runs instead of the repeats, and with only those pairs of runs that both lie in the validity

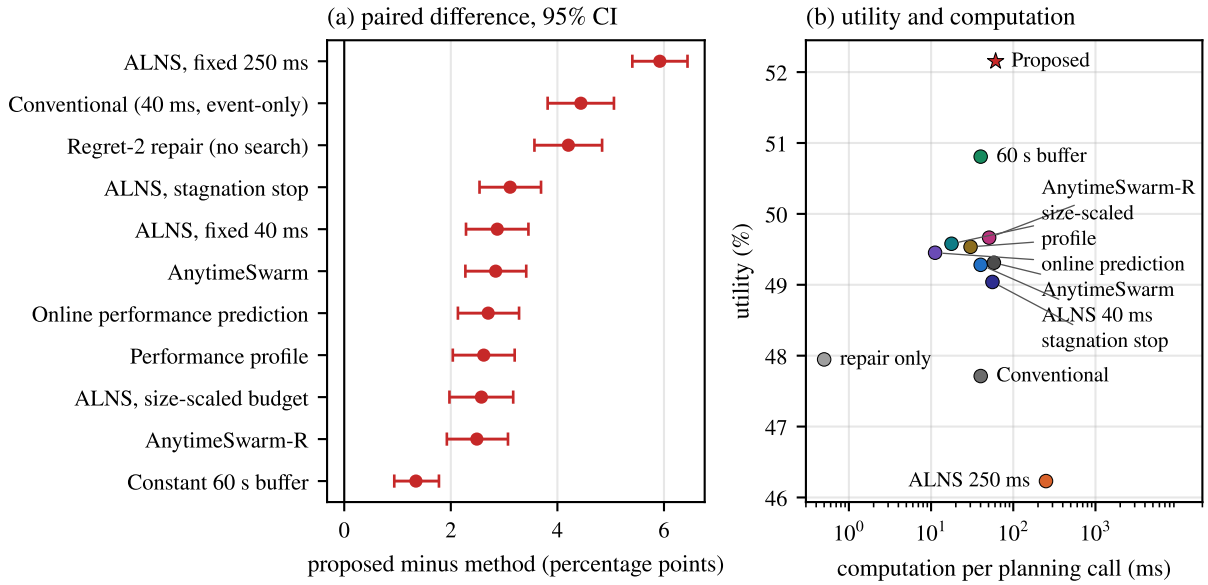


Figure 6: Study 2, tier B, held-out seeds 5000–5029 (48 600 runs): (a) utility of the proposed replanner minus each method, pooled over the 12 configurations, with 95% paired bootstrap intervals over 270 missions; (b) mean utility against mean computation per planning call. All methods except the conventional planner use the periodic trigger; ablations are omitted.

Table 9: Utility (%) per regime: disturbance level, time-window factor and latency factor κ_B (10, 50, 200), tier B held-out missions of study 2. Bold: best of the five methods.

	mid, windows ×1			mid, windows ×0.5			high, windows ×1			high, windows ×0.5		
Method	10	50	200	10	50	200	10	50	200	10	50	200
Proposed	**72.1**	**71.8**	**71.0**	**47.8**	**47.7**	**47.4**	**56.3**	**55.9**	**55.7**	33.5	33.3	33.3
Constant 60 s buffer	70.1	70.1	69.1	45.6	45.4	44.5	53.1	53.0	52.7	**35.7**	**35.5**	**34.9**
ALNS, fixed 40 ms	67.0	66.5	65.8	47.1	46.5	44.4	49.9	49.6	49.7	35.4	35.3	34.2
ALNS, fixed 250 ms	67.4	66.0	60.6	46.6	43.4	31.0	50.8	50.3	46.0	35.2	33.1	24.4
Conventional	65.6	65.5	63.9	45.4	44.9	42.7	48.7	48.6	47.5	34.1	33.6	32.2

range.

Table 10 summarises the results. The proposed replanner reached a utility of 59.89%, against 60.15% for ALNS with 40 ms, 57.26% for the constant buffer and 58.61% for the conventional planner. The paired differences were −0.25 [−4.02, +3.57] pp to ALNS with 40 ms and +1.28 [−2.20, +4.64] pp to the conventional planner, with Holm-adjusted p-values of 0.942 and 0.942. The fourth hypothesis is therefore not supported. The secondary comparison with the constant buffer gave +2.64 [−0.01, +5.58] pp. The sensitivity analyses lead to the same conclusion. With the original runs, the differences were −1.04 [−4.83, +2.77] pp and −0.02 [−3.66, +3.47] pp. With valid runs only, they were −2.02 [−6.25, +2.43] pp on 16 missions and −1.52 [−6.32, +3.37] pp on 14 missions. No interval in the three analyses excludes zero.

The mechanism seen in tier B reappears in the counts of late and skipped tasks. The proposed replanner served 0.90 tasks late per mission, against 1.20 for ALNS with 40 ms, which is 25% fewer. It skipped no task at departure, whereas ALNS with 40 ms skipped 0.23 and the conventional planner 0.57 tasks per mission. It computed 44.4 ms per call, against 40.2 ms for the fixed budget. The conventional planner called the search 6.3 times per mission instead of about 21.9, as in tier B.

The same 30 missions were also run in tier B with the rescaled parameters of tier C. As in study 1, tier B predicted the utility of a flight closely, with a correlation of $r = 0.86$ over 120 runs. On these missions,

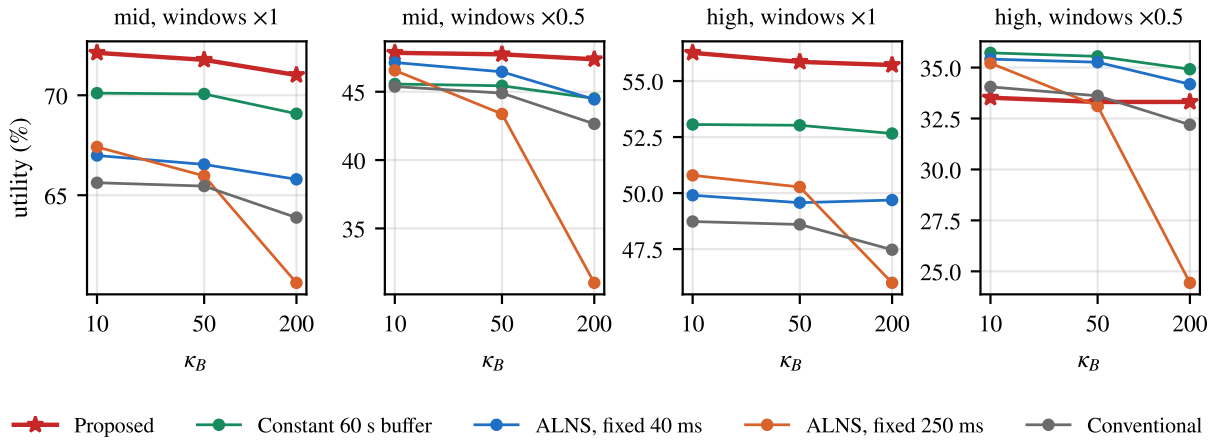


Figure 7: Study 2, tier B: utility per regime as a function of the latency factor κ_B, for mid and high disturbances with the original and with halved time windows (mean over 270 held-out missions per point). The proposed replanner is shown with stars.

Table 10: Tier C (PX4 SITL, Gazebo, ROS 2), study 2: 30 held-out missions per method, mid disturbances, $\kappa_B = 50$. Utility, tasks served late, tasks skipped at departure, planning calls and computation per call; tier B utility on the same rescaled missions.

Method	Utility C (%)	Late	Skipped	Calls	ms/call	Utility B (%)
Proposed	59.89	0.90	0.00	21.9	44.4	61.50
ALNS, fixed 40 ms	60.15	1.20	0.23	20.9	40.2	58.67
Constant 60 s buffer	57.26	0.83	0.00	21.0	40.2	59.60
Conventional (40 ms, event-only)	58.61	1.37	0.57	6.3	40.2	59.70

however, even tier B did not resolve the differences between the methods (Fig. 8). Its estimates were +2.82 [−1.20, +6.62] pp to ALNS with 40 ms, +1.90 [−0.59, +4.65] pp to the constant buffer and +1.80 [−2.31, +5.65] pp to the conventional planner. The tier C intervals overlap them. The per-mission difference to ALNS with 40 ms had a standard deviation of 10.9 pp in tier C. Confirming the tier B estimate of 2.8 pp on these missions with a power of 80% would require about 118 missions per method. Tier C thus shows that the proposed replanner runs on an autopilot and a physics engine with the same computation and fewer late tasks, but it cannot confirm the size of the gain measured in tier B.

7. Discussion and limitations

The two studies give an order of priorities for a multi-UAV replanner under uncertain execution. The planning model matters most. Moving every deadline by the predicted finish-time error raised the utility by more than any choice of stopping rule or trigger, and it removed the losses from late plans. The trigger comes second. On the nominal model of study 1, periodic replanning raised the utility of every method by more than the stopping rule did, but on the chance-constrained model its contribution fell to 0.22 pp, which suggests that both correct the same accumulated deviations, one before and one after they occur. The stopping rule comes last. It guards against long budgets on slow planners, where the full budget of 250 ms served 5.92 pp less reward than the proposed replanner and +11.34 [+10.78, +11.91] pp less at $\kappa_B = 200$, and it finds a suitable amount of computation on every fleet scale without tuning. On the proposed model, however, it was equivalent to a fixed budget of 40 ms that had been matched to its mean computation on development data.

The proposed model and the constant buffer both shift deadlines forward, and the buffer was the strongest baseline. The difference lies in the shape of the margin. The error grows with the lead time, so a constant margin is too large for tasks that are reached soon and too small for tasks at the end of a route. The

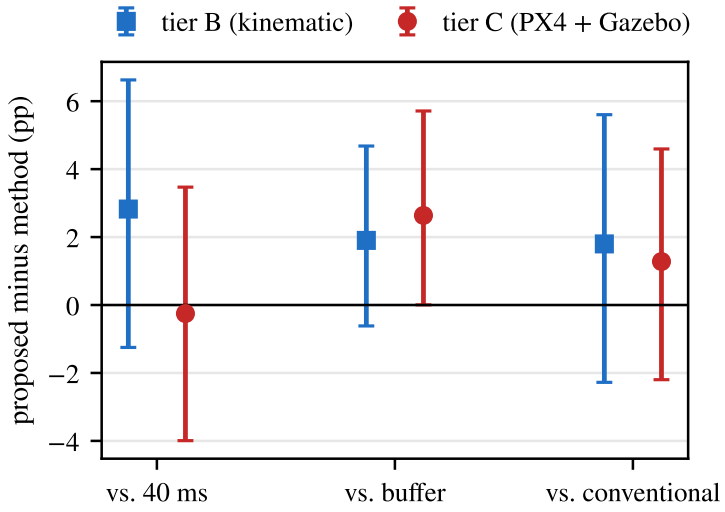


Figure 8: Study 2, tier B against tier C on the same 30 held-out missions (three UAVs, mid disturbances, $\kappa_B = 50$): paired difference of the proposed replanner minus each baseline with 95% bootstrap intervals.

proposed model scales the margin with the lead time and served more reward than the buffer in 9 of the 12 configurations. Under high disturbances with halved windows, the order reversed. A chance constraint discards every task whose probability of being served on time falls below $\Phi(z)$, even when the vehicle has nothing better to do. With short windows and large errors, many tasks fall below this threshold. An objective that weights each task by its probability of being served on time, instead of a hard constraint, would keep such tasks as fallback options and may remove this weakness.

The stopping rules from the metareasoning literature behaved like the learned rule. On the nominal model, the performance profile, online performance prediction, the learned predictor, the size-scaled budget and ALNS with 40 ms all lay within 0.38 pp of each other, although they used between 11.2 ms and 58.1 ms per call. The improvement curve of the ALNS used here flattens within a few tens of milliseconds, so every reasonable rule stops near the same point and the differences in reward stay small. Monitoring rules may matter more for search methods whose improvement continues for longer, or on slower hardware. The execution-aware delay cost of study 1 did not deliver the gain that motivated it either. It changed only the moment at which the search stops, and the remaining error was corrected at the next replanning call. Letting the error model shape the plan, as the proposed model does, was far more effective than letting it shape the decision to stop.

The two tiers play different roles. Tier B provides the number of missions that a paired comparison of replanners needs, and tier C checks that its conclusions survive an autopilot, a physics engine and a real middleware. In study 1, tier B predicted the utility of individual flights closely and reproduced the two robust findings, the loss of the long fixed budget and the lower computation of AnytimeSwarm-R. It did not predict the order of methods whose utilities differ by less than one percentage point, and neither tier can resolve such differences with the number of missions used here. The per-mission difference between two methods in tier C had a standard deviation of 7.7 pp, so confirming a difference of half a percentage point with a power of 80% would require about 1 800 missions per method, far beyond what a software-in-the-loop campaign of one night allows. A two-tier study should therefore state its effect sizes before running and plan the number of tier C missions from the smallest difference that it is meant to confirm. Study 2 repeated this pattern. Tier C reproduced the mechanism of the proposed model, fewer late and no skipped tasks, but not the utility gain, whose tier B estimate on the same missions was itself not significant. With a per-mission standard deviation

of 10.9 pp, about 118 missions per method would be needed.

Several limitations bound these conclusions. The finish-time error model is linear in the lead time and fitted per disturbance level, so the level itself is assumed to be known, and the confidence parameter z was selected per level on development missions. The chance constraint is imposed on every task separately rather than jointly on a route. Tier C uses three UAVs and fifteen tasks per mission, so it validates the decision problem at a small scale and with a limited number of missions. Software-in-the-loop simulation reproduces the autopilot and the vehicle dynamics but not the sensors, the radio link or the weather of a field test, and wind is constant during a mission. The latency factor emulates a slower planning computer instead of measuring one. All experiments ran on a single laptop, and the measured computation depends on it. Because the stopping rules read a wall clock, repeated runs are not bit-identical, and the mean utility of a method varied by up to 0.12 pp between two runs of the same missions. Finally, all methods share one ALNS and one family of scenarios, and other search methods or mission types may change the size of the gains.

8. Conclusion

This paper proposed AnytimeSwarm-CC, a multi-UAV replanner that plans against a chance constraint on the finish time of every task, replans on events and periodically, and stops an anytime ALNS with a learned rule. Because the calibrated finish-time error grows linearly with the lead time, the chance constraint reduces exactly to an effective deadline per task, so the proposed model costs no more to solve than the nominal one. The replanner was evaluated in two pre-registered studies with a kinematic tier and a PX4 software-in-the-loop tier on held-out missions.

In the kinematic tier, the proposed replanner served more reward than ALNS with fixed and size-scaled budgets, than stopping rules based on performance profiles and on online performance prediction, than an execution-aware stopping rule on the nominal model and than a robust constant buffer, by 1.35 to 2.87 percentage points, and 9.3% more than a conventional event-triggered planner. Ablations showed that the gain comes from the chance-constrained model. On this model, the learned stopping rule was equivalent to a well-chosen fixed budget, and its value lies in avoiding the losses of long budgets on slow planners and the tuning of a budget per fleet scale. Under high disturbances with halved time windows, the proposed model was too cautious. In software-in-the-loop flights with PX4 and Gazebo, the proposed replanner served fewer tasks late at the same computation, but its utility did not differ detectably from the baselines on 30 missions.

Future work will replace the hard chance constraint by an objective that weights every task by its probability of being served on time, adapt the confidence parameter online, and run the replanner on embedded hardware and in field tests.

CRediT authorship contribution statement

Cao-Phuc Ha: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. Anh Nguyen Vu Minh: Software, Validation, Writing – review and editing. Phu-Nguyen Le: Supervision, Methodology, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The planner, both simulation tiers, the experiment configurations with their SHA-256 hashes, the pre-registered protocol, all raw results and the scripts that produce every number, table and figure of this paper will be made available in a public repository upon publication.

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